

Tonghe Zhuang

Data Scientist & AI Engineer — Multi-modal ML · Signal Processing · Health Data
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SUMMARY

PhD scientist with three years of postdoctoral research building multi-modal models and the pipelines around them. I train deep networks in PyTorch on noisy, expensive, high-dimensional data — video-rate neural signals, imaging, and over a million human behavioural judgements — and I have spent my career on the question of what a model has actually learned versus what its metrics suggest. Used to owning a problem end to end, from data collection through modelling to a result other people can act on, in small teams where nobody else was going to do it.

TECHNICAL SKILLS

Languages & tools Python (primary), Git, AWS, Jupyter; MATLAB; R (basic)

Machine learning PyTorch; CNN architectures (ResNet, AlexNet, EEGNet); multi-modal representation learning across text, image and signal inputs; encoding and decoding models; transfer learning; cross-validated benchmarking and model comparison

Signal processing EEG acquisition, preprocessing and artefact handling; feature extraction and classification from noisy time-series; time-resolved decoding; MRI preprocessing pipelines

Data & statistics pandas, NumPy, SciPy, scikit-learn; large-scale data cleaning and quality control (1M+ records); experimental design and A/B-style comparison; statistical inference and significance testing

Domain Health and clinical research data; multivariate pattern analysis (MVPA); representational similarity analysis (RSA)

EXPERIENCE

Postdoctoral Researcher — Computational Cognitive Neuroscience

Max Planck Institute for Human Cognitive and Brain Sciences & Justus Liebig University Giessen | Nov 2023 – Sep 2026 | Germany

- Built multi-modal models of how the same concept is represented when it arrives as text versus as an image; trained and benchmarked deep networks in PyTorch against a dataset of over one million human judgements, and quantified where model and human representations diverge.
- Designed and shipped end-to-end Python pipelines across behavioural, fMRI and EEG data — ingestion, quality control, feature extraction, model fitting and statistical validation.
- Built and documented a large-scale imaging dataset for reuse: acquisition protocol, preprocessing pipeline, quality control and documentation.
- Wrote a preprint on AI-assisted coding in scientific research.
- Mentored one MSc thesis student and 12 interns; ran an internal training course on representational analysis methods.

Doctoral Researcher — Cognitive Neuroscience

University of Regensburg | Nov 2018 – Oct 2023 | Germany

- Ran one EEG, two fMRI and eleven behavioural studies end to end: experimental design, acquisition from 200 participants, preprocessing, and multivariate analysis.
- Developed time-resolved decoding methods for EEG — artefact handling, feature extraction and multivariate classification across the time course of a signal — recovering hierarchical structure in observed actions. Published in *Journal of Neuroscience* (2023).
- Delivered two first-author publications and secured two competitive funding awards during the PhD.

Concurrent roles

- International Research Associate, Joint International Research Unit on Neuroplasticity (Laval University, Canada / University of Liège, Belgium) — clinical research collaboration on disorders of consciousness.

- Associate Editor, Brain Connectivity — peer review and editorial decisions.
- Research Mentor, Neuromatch Academy — mentoring international teams through applied computational projects.

EDUCATION

PhD, Cognitive Neuroscience — University of Regensburg, Germany | 2018–2023

Coursework: Deep and Machine Learning for Neuroscience and Medicine; EEG Analysis

MSc, Cognitive Neuroscience — Beijing Normal University, China | 2015–2018

Key Laboratory of Cognitive Neuroscience and Learning / IDG-McGovern Institute for Brain Research

BSc, Psychology — Liaoning Normal University, China | 2011–2015

SELECTED PUBLICATIONS

- Zhuang, T., Kabulska, Z., & Lingnau, A. (2023). The representation of observed actions at the subordinate, basic and superordinate level. *Journal of Neuroscience*.
- Kabulska, Z., Zhuang, T., & Lingnau, A. (2024). Overlapping representations of observed actions and action-related features. *Human Brain Mapping*, 45(3).
- Zhuang, T., & Lingnau, A. (2021). The characterization of actions at the superordinate, basic and subordinate level. *Psychological Research*.
- Wang, X.*, Zhuang, T.*, Shen, J., & Bi, Y. (2018). Disentangling representations of shape and action components in the tool network. *Neuropsychologia*, 117. *co-first author
- Zhuang, T., & Lin, Z. (2024). The why, what, and how of AI-based coding in scientific research. Preprint.

ADDITIONAL

Languages Chinese (native), English (fluent), German (A2, in progress)

Awards Research Scholarship for Female Early Career Researchers (2021–2022); China Scholarship Council Scholarship (2018–2021); Student Abstract Award, CAOs Workshop (2017)